Nicolo DiFerdinando

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About Me

AI/ML and data engineer who ships production systems end to end and measures them with numbers. Builds LLM-powered automation, dbt and Snowflake data pipelines, a semantic layer served over MCP, and calibrated ML models. Cal Poly Industrial Engineering graduate, with security automation and self-hosted infrastructure as supporting depth.

Education

Bachelor of Science in Industrial Engineering *Sept 2021 – June 2026*
California Polytechnic State University, San Luis Obispo
Coursework: Data Science, Computer Science, Data Analytics, Enterprise Analytics, Database Management & System Design, Operations Research, Machine Learning, Statistics & Design of Experiments, Simulation, Human Factors & Ergonomics, Usability / HCI, Project Management, Internal Systems Management, RAG-assisted LLM systems.

Experience

- **AI-Driven Solutions Analyst (Full-time)** – *PeopleConnect* *Aug 2026 – Present*
  - **AI Automation & Tooling:** Build and deploy LLM-powered tools and Python scripts using OpenAI, Claude, Zapier, and Airflow to automate high-impact workflows across Growth, BI, and Operations.
  - **Growth & Analytics Enablement:** Partner with Marketing and Finance on creative generation and campaign optimization, applying ML and prompt-engineered systems to improve forecasting across Snowflake, dbt, and Sigma.
  - **Cross-Functional Impact:** Translate business problems into AI-driven solutions. Run structured experiments and measure outcomes on time savings, revenue lift, and cost efficiency to scale what works.
- **Business Intelligence Intern (Part-time)** – *PeopleConnect* *Oct 2025 – June 2026*
  - **Dashboard Development:** Build interactive dashboards using Sigma Computing, Tableau, and Snowflake serving 50+ stakeholders across Growth Operations, Executives and Finance through cross-functional collaboration.
  - **Data Pipeline Engineering:** Develop and monitor data pipelines using dbt core/fusion semantic layers with CI/CD workflows; manage ETL pipelines for incremental logic with code review processes across the data engineering team.
  - **Semantic Layer Service:** Built a service embedding company metrics, dimensions, and entities over the dbt semantic layer, exposing governed definitions through MCP connectors (Claude, ChatGPT), a Slack bot, and more.
  - **Reporting Migration:** Assisted in migrating enterprise reporting from Tableau to Sigma Computing, streamlining dashboard delivery and reducing licensing overhead.
- **Cybersecurity & IT Intern (Full-time)** – *PeopleConnect* *June 2025 – Sept 2025*
  - **SOC 2 Compliance:** Developed bash scripts automating audit reporting workflows, completing SOC 2 audit in 4 weeks vs. projected 10; communicated findings to engineering and compliance stakeholders.
  - **Enterprise Security:** Deployed vulnerability scanning and testing solution for 200+ endpoints; created reusable automation scripts allowing further visibility into on prem devices, reducing risk surface.
- **Assistant Student SOC Analyst (Part-time)** – *Cal Poly ITS* *April 2024 – June 2025*
  - **Security Automation & Incident Response:** Built a Python Slack app automating SOC workflows (email confirmation, user suppression, blocking) by integrating Splunk SOAR playbooks with Slack Bolt REST APIs for real-time incident response, and analyzed and responded to 50+ weekly security incidents (suspicious logins, phishing, anomalous network traffic) using Splunk Enterprise, collaborating with IT teams on remediation.
- **CyberSecurity Intern (Full-time)** – *loanDepot* *Summer 2022*
  - Designed and deployed phishing simulation campaigns to educate employees on social engineering threats. Gained hands-on exposure to incident response procedures, compliance auditing, and security architecture.

Projects

- **Zedi – Personal Finance SaaS:** Shipped a full-stack finance platform with a Rust/Axum API, Next.js/TypeScript frontend, PostgreSQL with row-level multi-tenancy, Claude Vision receipt OCR, and Stripe billing. (zedi.dev)
- **BBLMLP – Baseball ML Prediction:** Statcast ingest into DuckDB, point-in-time features with leakage tests, walk-forward backtests and Brier/reliability calibration; a 139-feature LightGBM ties a tuned Elo baseline.
- **Flora – Senior Capstone (Bloomfilter.ai):** Year-long senior group project working with Bloomfilter improving HCI design for AI agent integration in project management tools, focused on increasing user adoption in enterprise software.
- **HomeLab Infrastructure:** Single-node Proxmox lab running Tailscale, n8n automation, Telegraf into ClickHouse, Proxmox Backup Server, llama.cpp OCR on a GPU container, and a self-hosted Zedi; configs and runbooks in git.

Leadership

- **Cal Poly Club Baseball** – Co-President (June 2025 – June 2026), Player (2023 – June 2026)
- **Kappa Sigma Fraternity** – President (Dec 2022 – Mar 2023), Risk Manager (Sept 2021 – June 2025)

Skills

Languages: Python, SQL, Rust, TypeScript, JavaScript, Swift/SwiftUI, Bash, YAML, HTML/CSS
Technologies: Node.js, React, Next.js, Streamlit, PyTorch, LightGBM, DuckDB, LangChain, PostgreSQL, MongoDB (NoSQL), Firebase, Supabase, Docker, Proxmox, Terraform, Git, CI/CD, Linux, Splunk, Splunk SOAR, Snowflake, dbt, Tableau, Sigma Computing, ClickHouse, Grafana, GCP, Cloudflare, Railway, Stripe, MCP